CYBERSECURITY HOMELAB

Enterprise-Grade Network Security Lab

Design, Implementation & Analysis Report

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Computer Networking

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Executive Summary

This report presents the design, deployment, and evaluation of a comprehensive cybersecurity home laboratory that accurately replicates a small-to-medium enterprise (SME) network. Built on VMware Workstation Pro 17 running on a Dell XPS 15 9570 with 32 GB RAM, the lab integrates industry-standard security technologies including pfSense, Active Directory, Security Onion, Splunk SIEM, and Wazuh HIDS.

The project demonstrates end-to-end competency across network segmentation, identity management, intrusion detection, log analysis, and vulnerability management — skills directly aligned with roles such as SOC Analyst, Network Security Engineer, and Security Operations Specialist.

Key Highlights

- Multi-segment network with zero-trust firewall policies (pfSense)
- Active Directory domain with fine-grained Group Policy enforcement
- Full SIEM pipeline: log ingestion → correlation → dashboards (Splunk + Wazuh)
- Network Security Monitoring with Security Onion IDS/IPS
- Vulnerability scanning workflow using Nessus
- DMZ hosting web and mail servers with SSL/TLS management
- Structured 5-phase implementation methodology

1. Introduction

Theoretical knowledge alone is insufficient preparation for a career in cybersecurity. Hands-on experience with enterprise tools, network architectures, and realistic attack-defence scenarios is essential. This homelab was created to bridge that gap — providing a controlled, repeatable environment to practise offensive and defensive security skills without risk to production systems.

The lab replicates an SME network topology, complete with internal workstations, a domain controller, DMZ services, dedicated security monitoring hosts, and a multi-segment firewall. Every architectural decision was guided by industry frameworks including NIST SP 800-53 and the CIS Controls.

2. Project Objectives

The following objectives governed the design and implementation phases:

- Construct a virtualised environment that faithfully mirrors an SME production network.
- Deploy and configure core network services: DHCP, DNS, and Active Directory Domain Services (AD DS).
- Implement a multi-layer security stack: next-generation firewall, IDS/IPS, and SIEM.
- Harden all endpoints and servers against common attack vectors.
- Establish centralised log collection, correlation, and real-time alerting.
- Create a repeatable platform for ongoing red/blue team exercises and skill development.

3. Network Architecture & Design

The lab is divided into four isolated network segments, each with a defined security boundary enforced by pfSense. This defence-in-depth model ensures that a compromise in one zone cannot freely propagate to others.

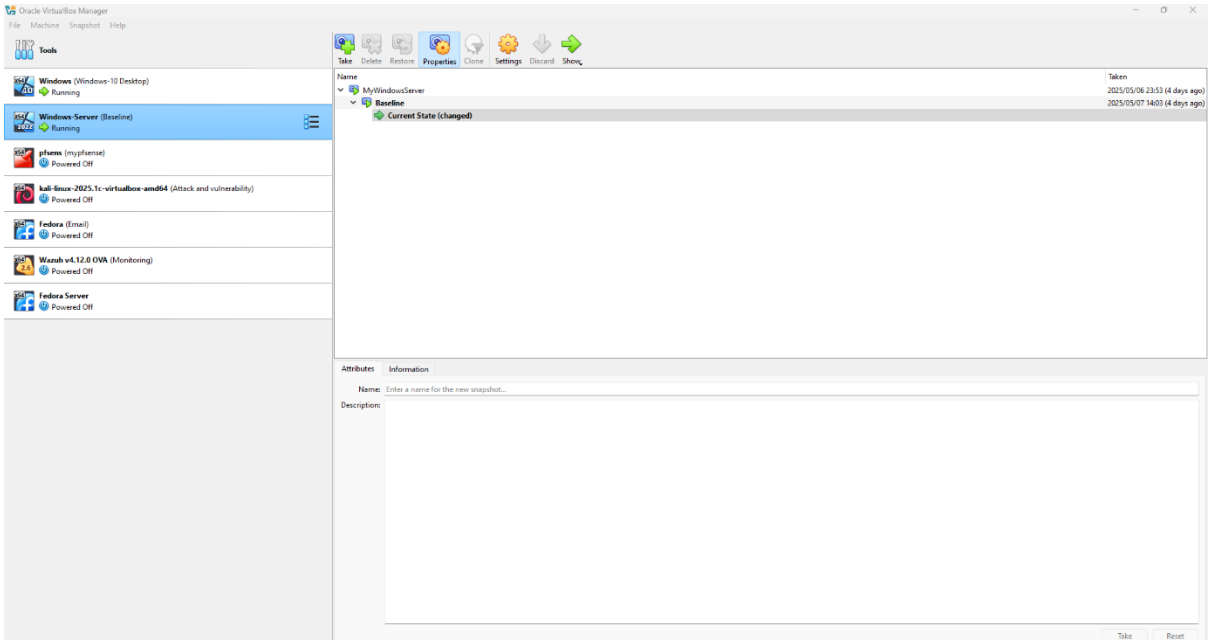


Figure: Lab network topology — overview

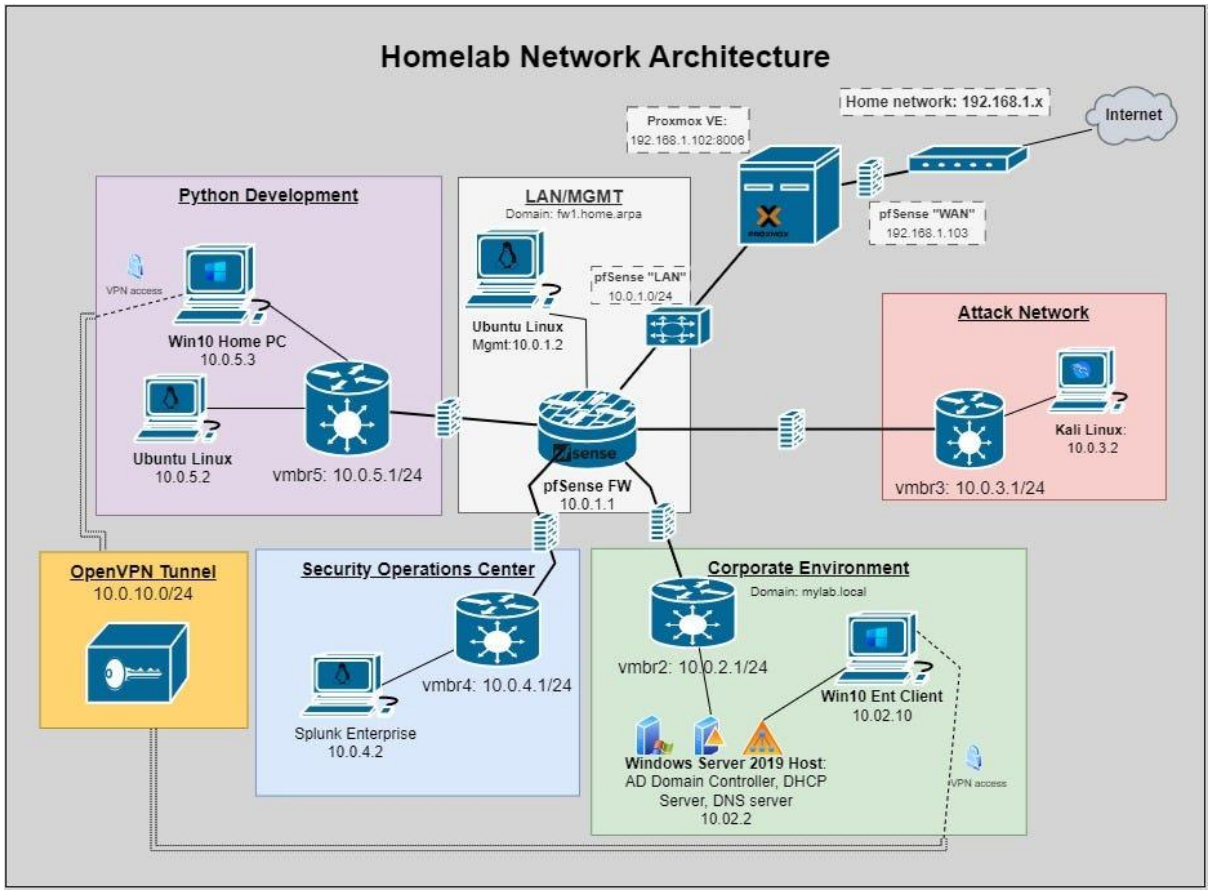


Figure: Detailed network topology diagram

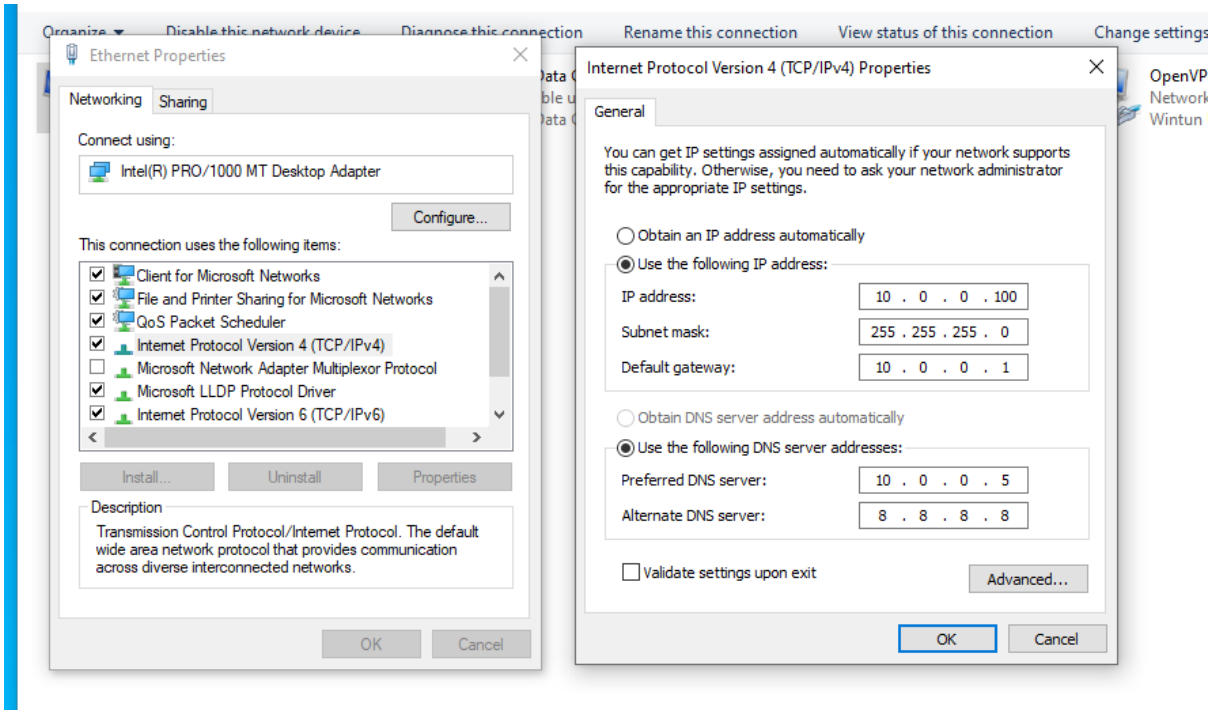


Figure: Network segmentation — Windows client and server segments

Zone	CIDR	Gateway	Purpose
Management	10.10.10.0/24	10.10.10.1	Administrative & out-of-band access
Corporate LAN	192.168.1.0/24	192.168.1.1	Internal users & workstations
DMZ	172.16.1.0/24	172.16.1.1	Public-facing web & mail servers
Security	10.0.0.0/24	10.0.0.1	SIEM, IDS/IPS & monitoring tools

Inter-zone traffic is governed by explicit allow rules built on zero-trust principles: all traffic is denied by default, and only the minimum required communications are permitted. NAT and traffic shaping are applied at the WAN boundary to manage bandwidth and mask internal addressing.

4. Hardware Specifications

Component	Specification
Host System	Dell XPS 15 9570
Processor	Intel Core i7-8750H — 6 cores / 12 threads @ 2.2–4.1 GHz
RAM	32 GB DDR4 — supports 10+ concurrent VMs
Primary Storage	1 TB NVMe SSD (OS + active VMs)
Secondary Storage	2 TB External USB 3.1 SSD (VM library & backups)
Networking	1 Gbps Ethernet + Wi-Fi (dual connectivity)
Hypervisor	VMware Workstation Pro 17 (nested virtualisation enabled)

5. Implemented Services

5.1 pfSense Next-Generation Firewall

Platform: FreeBSD-based pfSense 2.7.0

Role: Network gateway, NAT, DHCP, VPN, traffic shaping

pfSense anchors the entire lab network. It separates the four segments, enforces firewall policies, and provides centralised DHCP services. Key configuration decisions include:

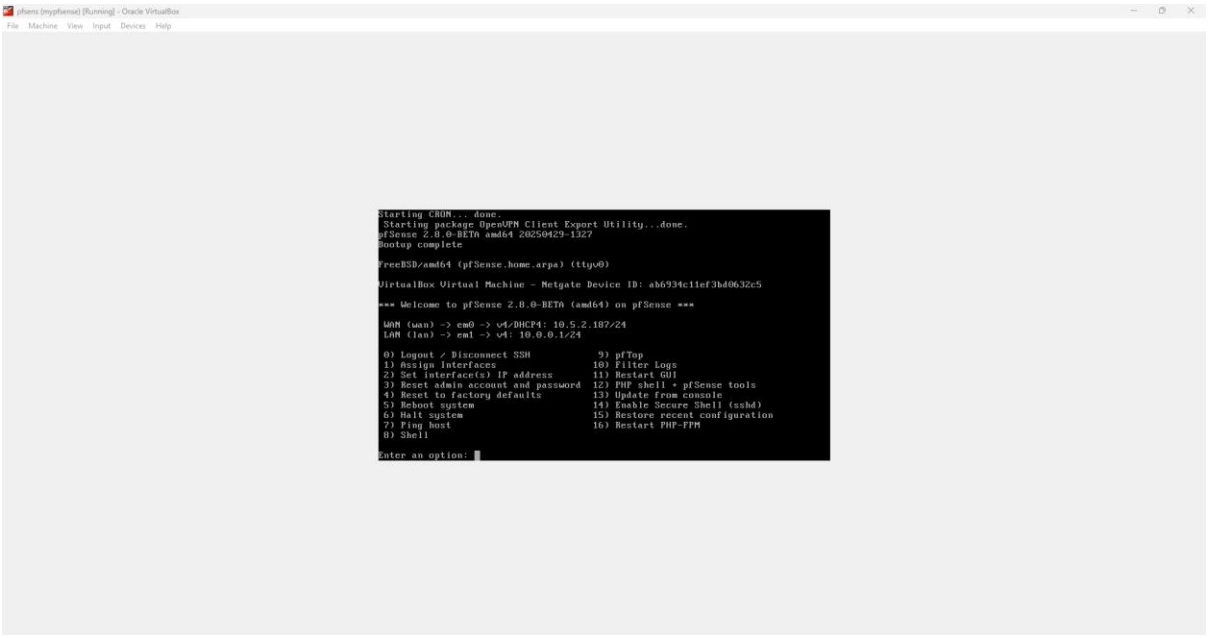


Figure: pfSense dashboard — configuration overview

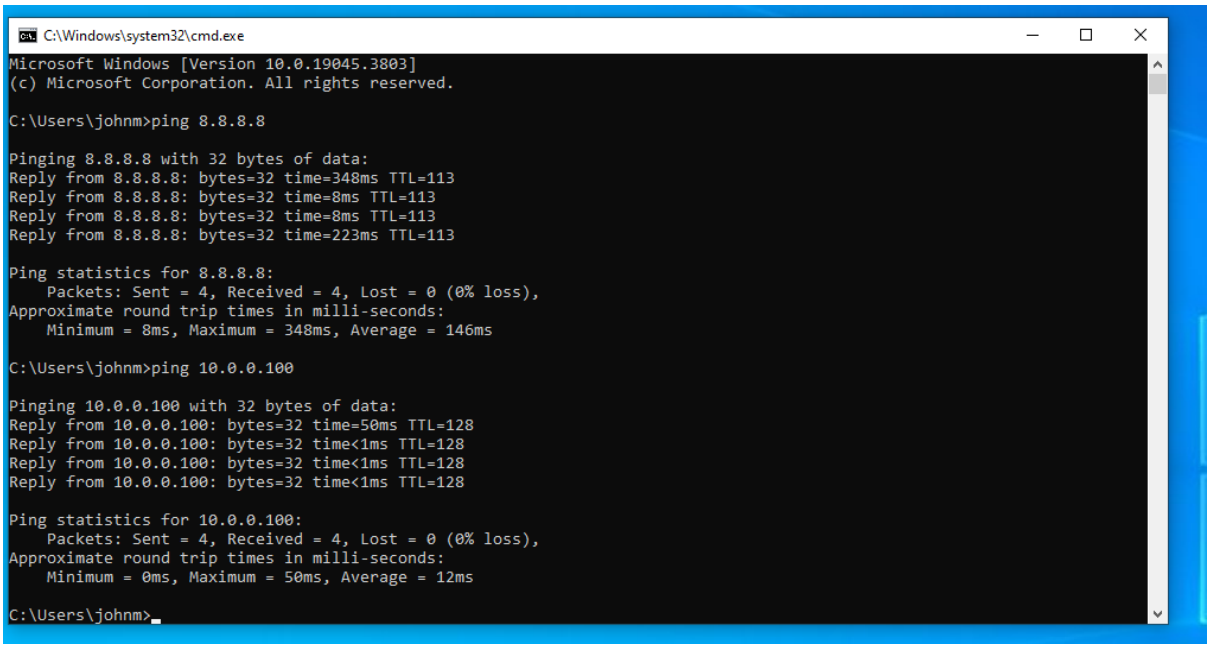


Figure: Connectivity and IP address verification on Windows client

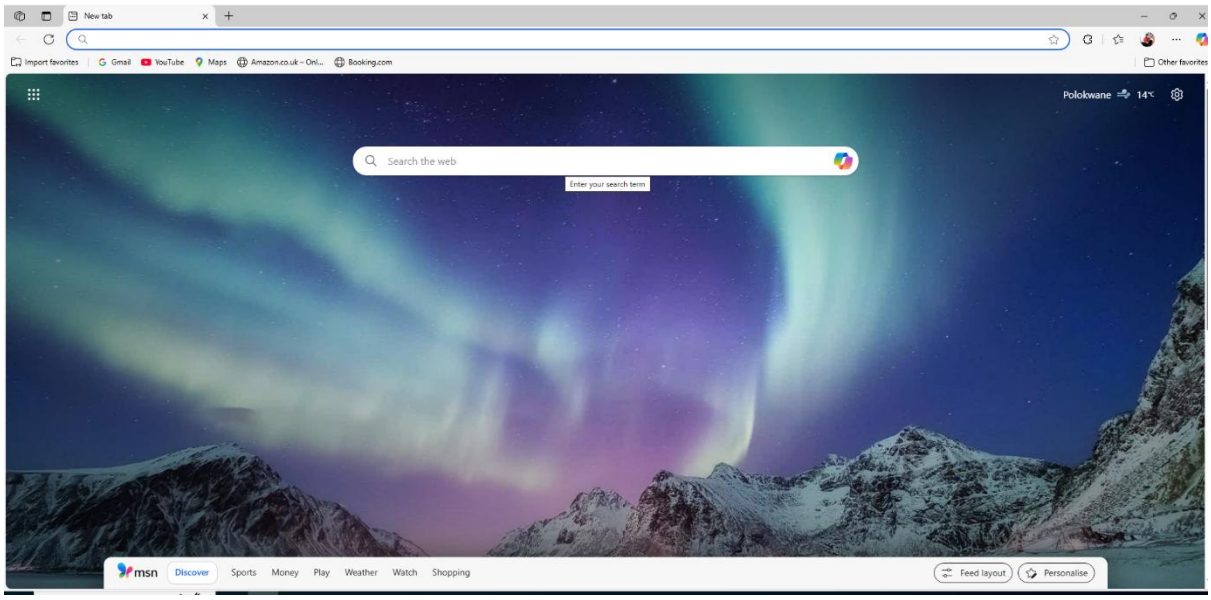


Figure: Windows client — internet connection confirmed

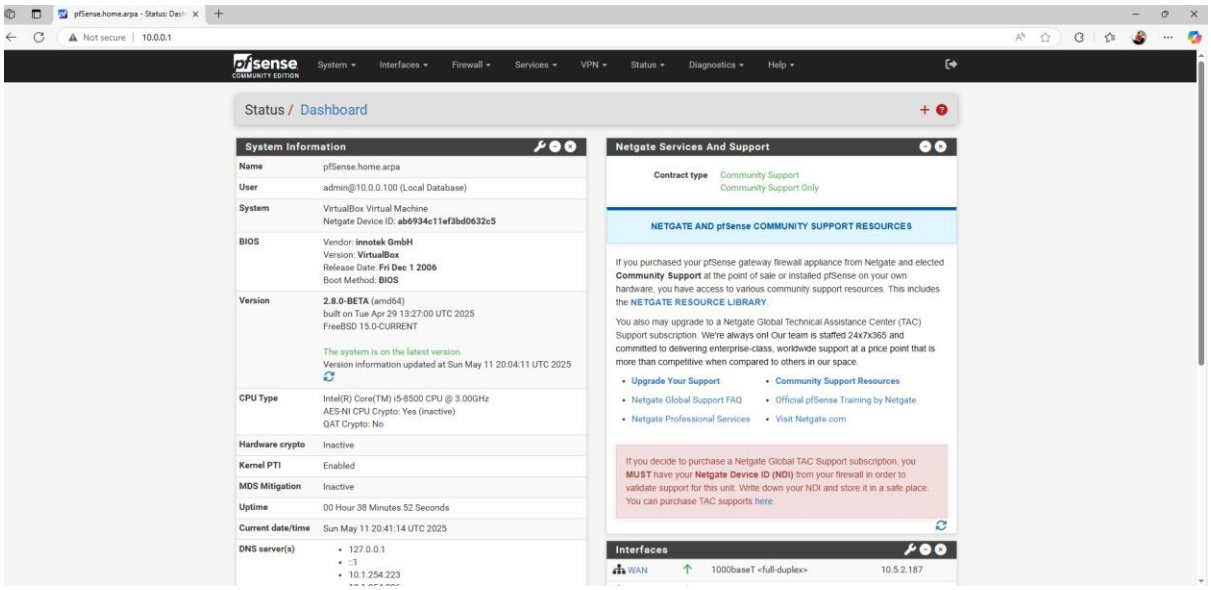


Figure: pfSense admin interface from Windows client

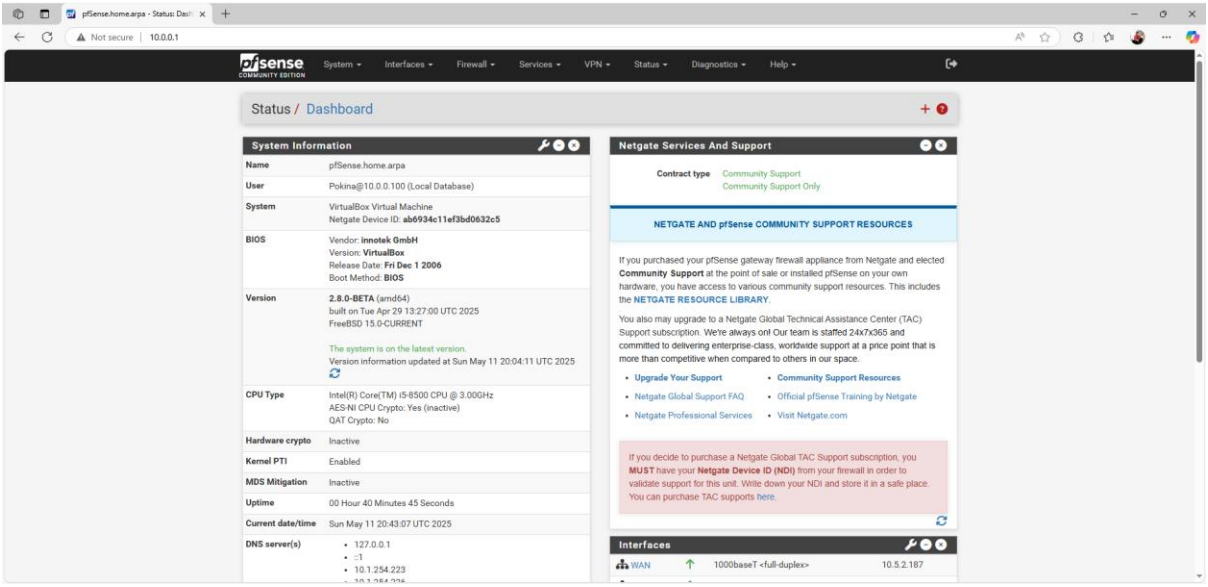


Figure: Successful client login via pfSense

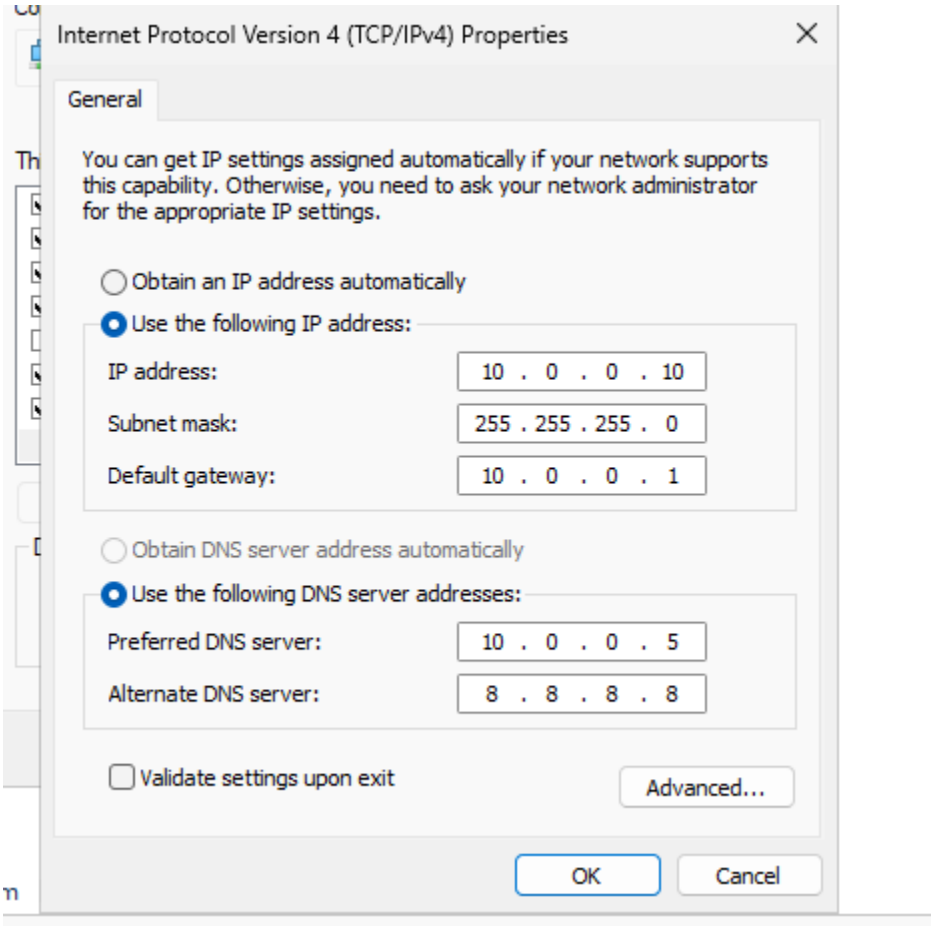


Figure: IP addressing and default gateway configuration

- Zero-trust default-deny rule base with explicit allow statements for required flows.
- OpenVPN server enabling secure remote access to the lab from external machines.
- URL filtering and content control to simulate enterprise web proxy capabilities.
- Traffic shaping to emulate real-world WAN bandwidth constraints.

5.2 Windows Server 2019 — Domain Controller

Services: AD DS, DNS, DHCP failover, File Services, Group Policy

Domain: lab.local (structured OU hierarchy)

Active Directory provides centralised identity management for all Windows hosts in the Corporate LAN segment. The domain is configured with:

- Organisational Unit (OU) structure mirroring a real business (IT, Finance, HR, Servers).
- Group Policy Objects enforcing password complexity, account lockout, and audit policies.
- Integrated DNS resolving both internal and external names.
- File server with permission-based share access demonstrating the principle of least privilege.

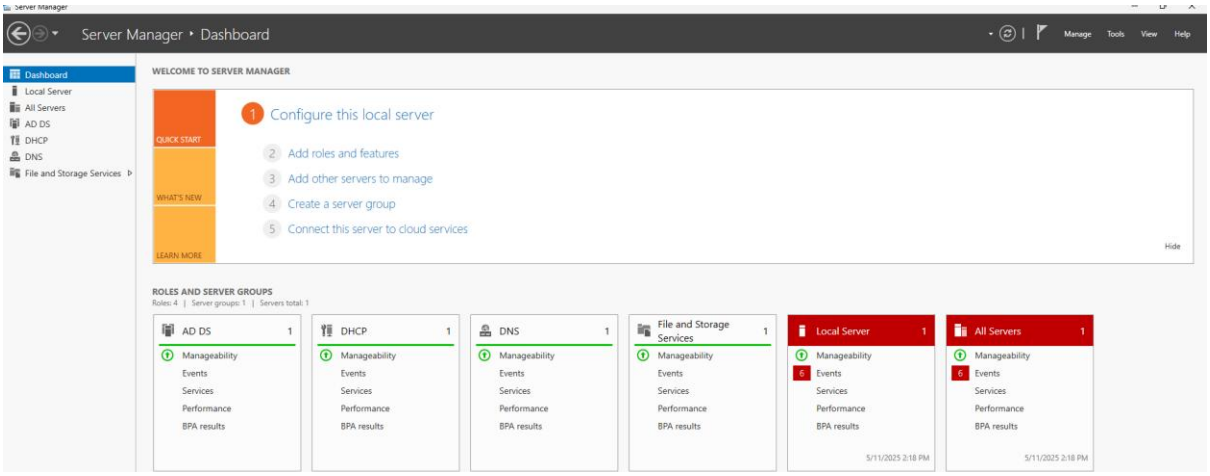


Figure: Windows Server — server manager and services

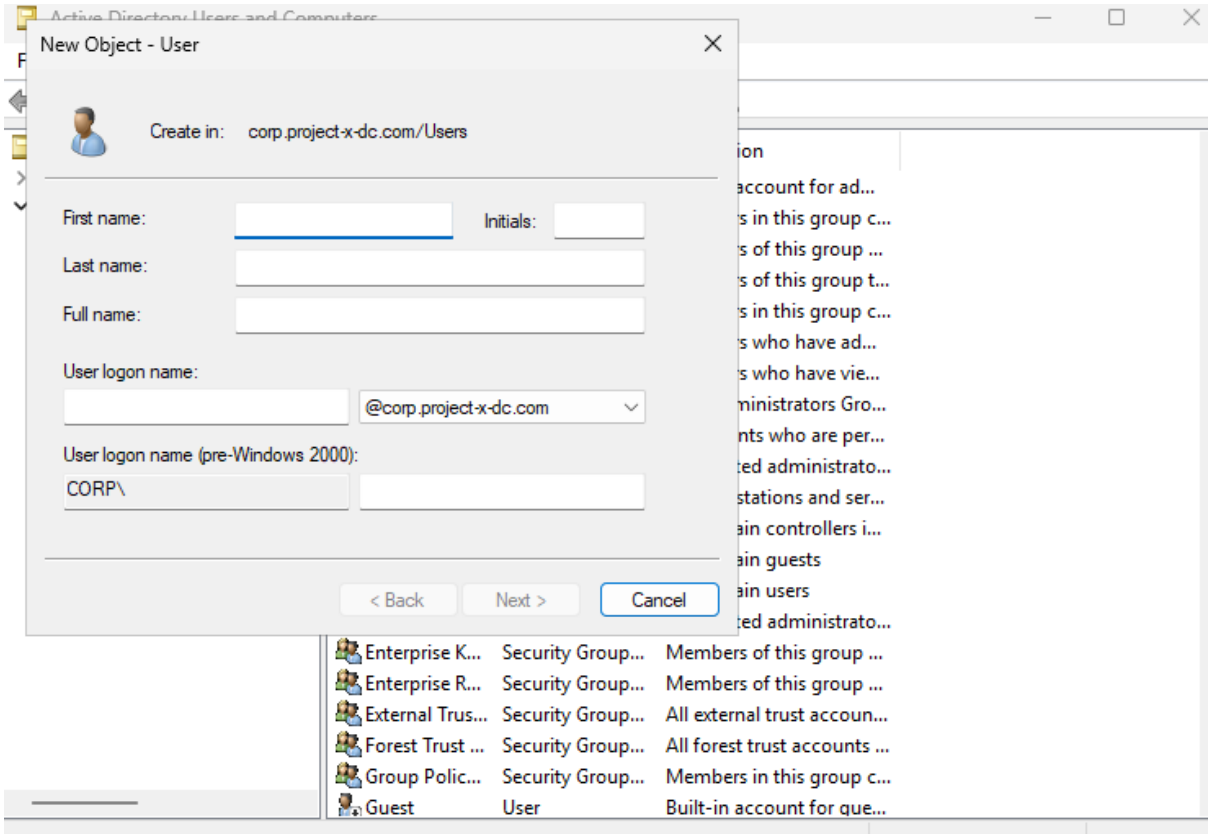


Figure: Active Directory — user accounts in AD DS

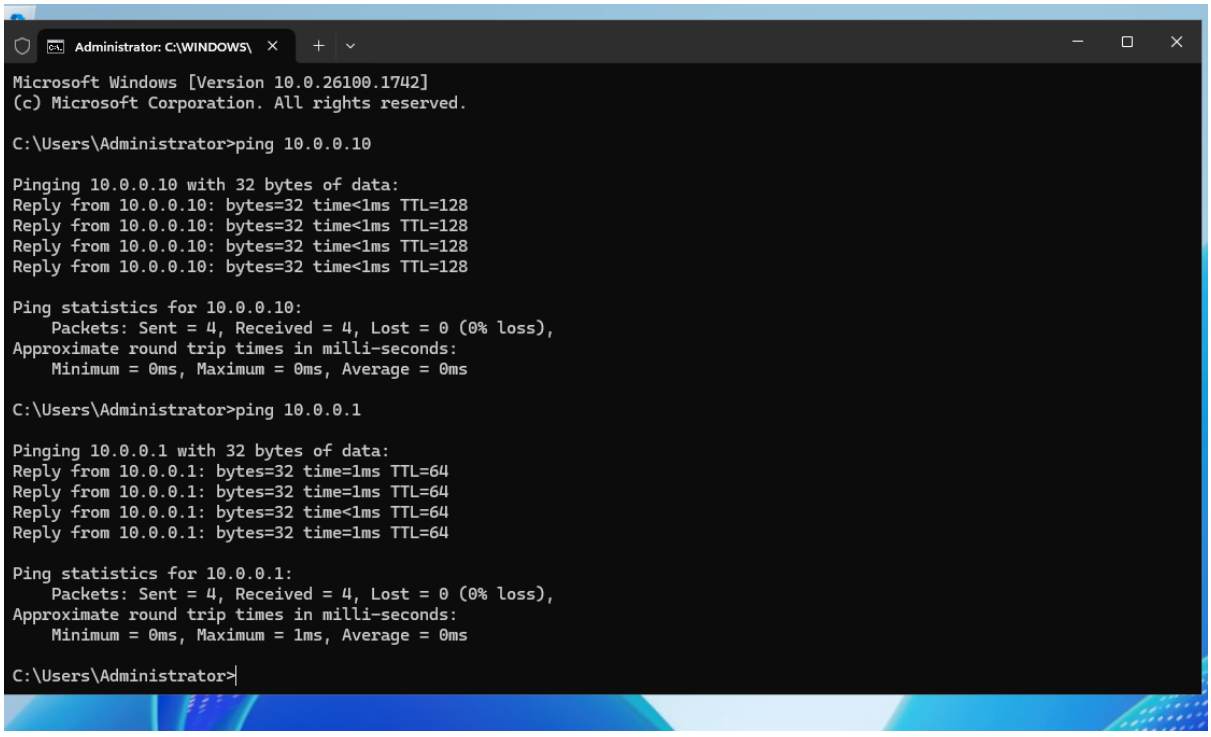


Figure: Windows Server — internet connectivity confirmed

5.3 Security Onion — Network Security Monitoring

Version: Security Onion 2.4

Role: IDS/IPS, full-packet capture, network traffic analysis

Security Onion sits on the dedicated Security VLAN and receives a copy of all inter-zone traffic via a SPAN port. It runs Suricata for signature-based detection and Zeek for protocol analysis, feeding alerts into its built-in Elastic Stack for investigation.

- Custom Suricata rule sets tuned to the lab environment.
- Zeek scripts generating structured connection, DNS, and HTTP logs.
- Hunt-mode dashboards for proactive threat hunting exercises.

5.4 Splunk Enterprise — SIEM

Version: Splunk Enterprise 9.0.4

Role: Log aggregation, correlation, alerting, dashboards

Splunk serves as the central SIEM, ingesting logs from all hosts via Universal Forwarders and Syslog. Custom dashboards surface authentication events, network anomalies, and policy violations.

- Correlation searches detecting brute-force, lateral movement, and data exfiltration patterns.
- Real-time alerts with configurable thresholds and notification actions.
- Role-based dashboards for SOC analyst and management-level views.

5.5 Wazuh — Host-Based Intrusion Detection

Role: HIDS, file integrity monitoring, compliance checks

Wazuh agents are deployed on all endpoints, providing host-level visibility complementing Security Onion's network-level monitoring. Wazuh monitors file integrity, registry changes, and running processes, generating alerts for suspicious host activity.

5.6 Kali Linux — Adversary Simulation

Version: Kali Linux 2023.1

Role: Penetration testing, red team exercises

Kali Linux is isolated on the Security VLAN and used to conduct controlled penetration tests against lab targets. Attack exercises validate the effectiveness of firewall rules, IDS signatures, and SIEM correlation logic.

5.7 Nessus — Vulnerability Scanner

Role: Credentialed vulnerability assessments

Nessus performs authenticated scans against all hosts on a scheduled basis, producing prioritised vulnerability reports. Results are tracked over time to demonstrate the effectiveness of the patch management process.

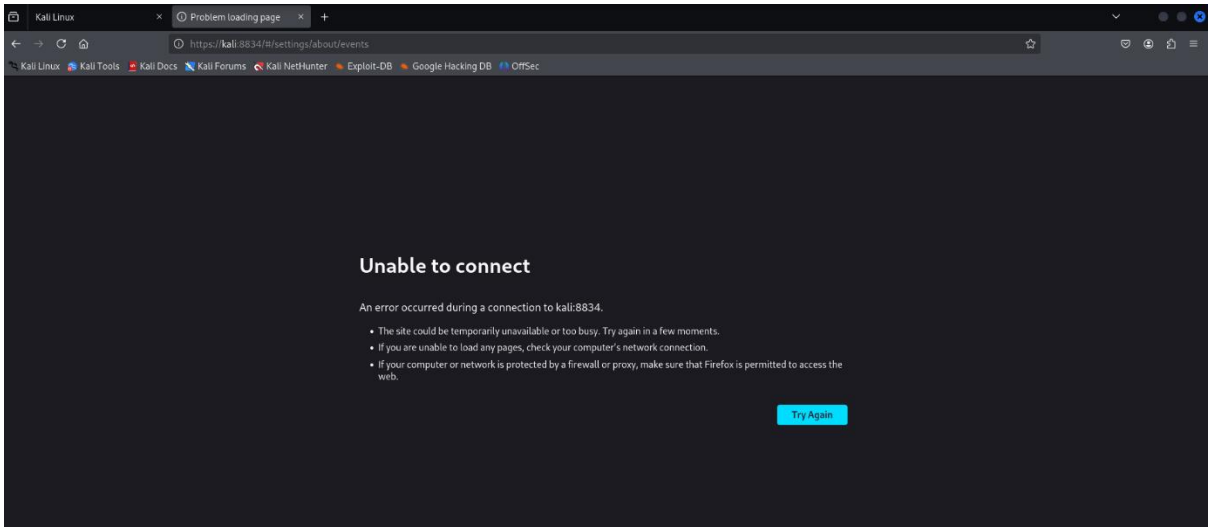


Figure: Nessus — initial connection issue

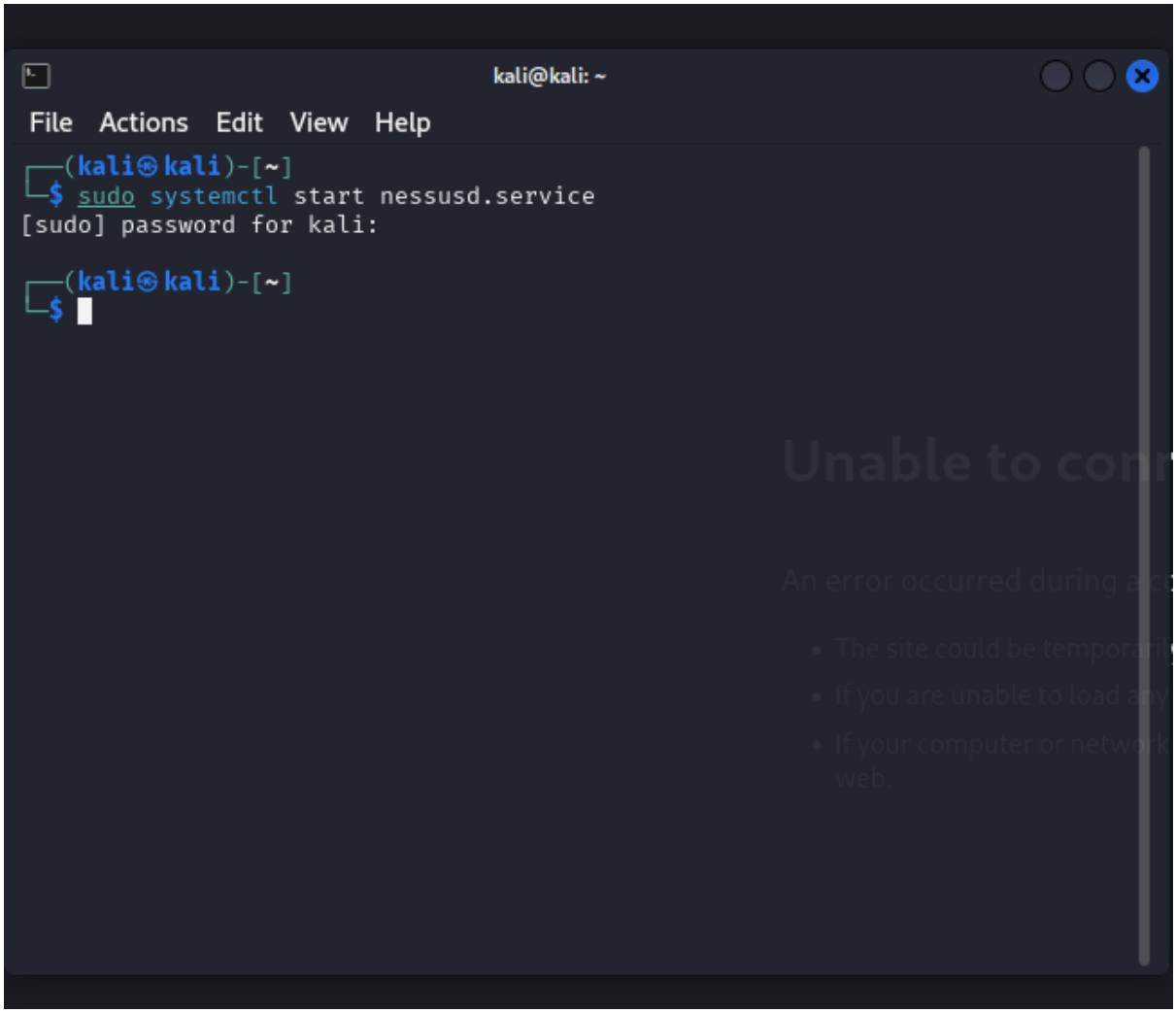


Figure: Nessus — connected successfully via terminal

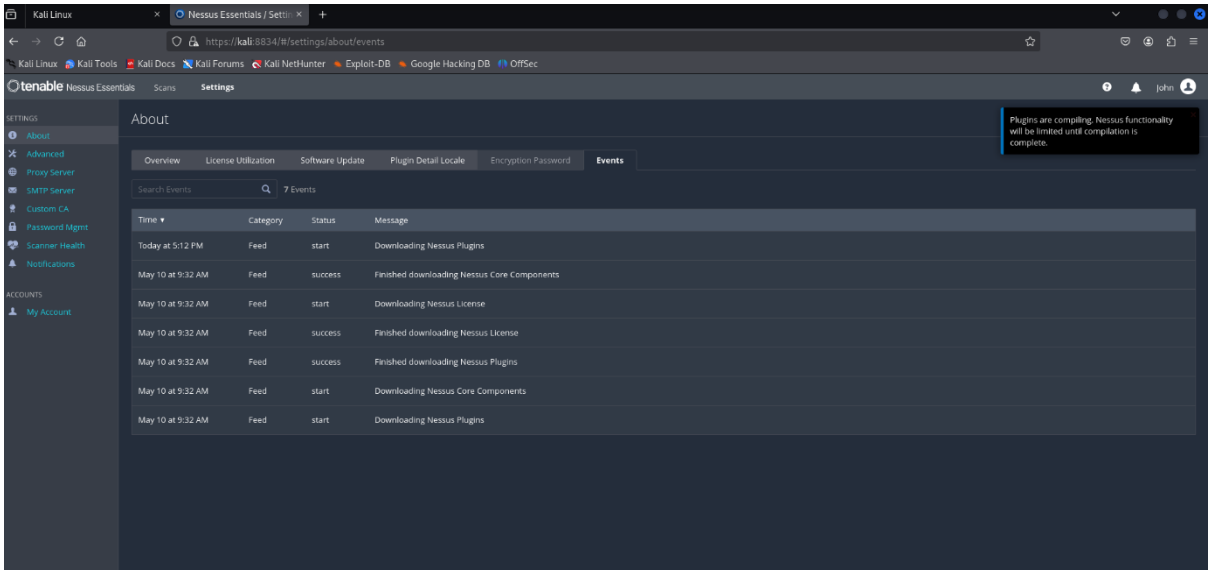


Figure: Nessus — new scan configuration pending resolution

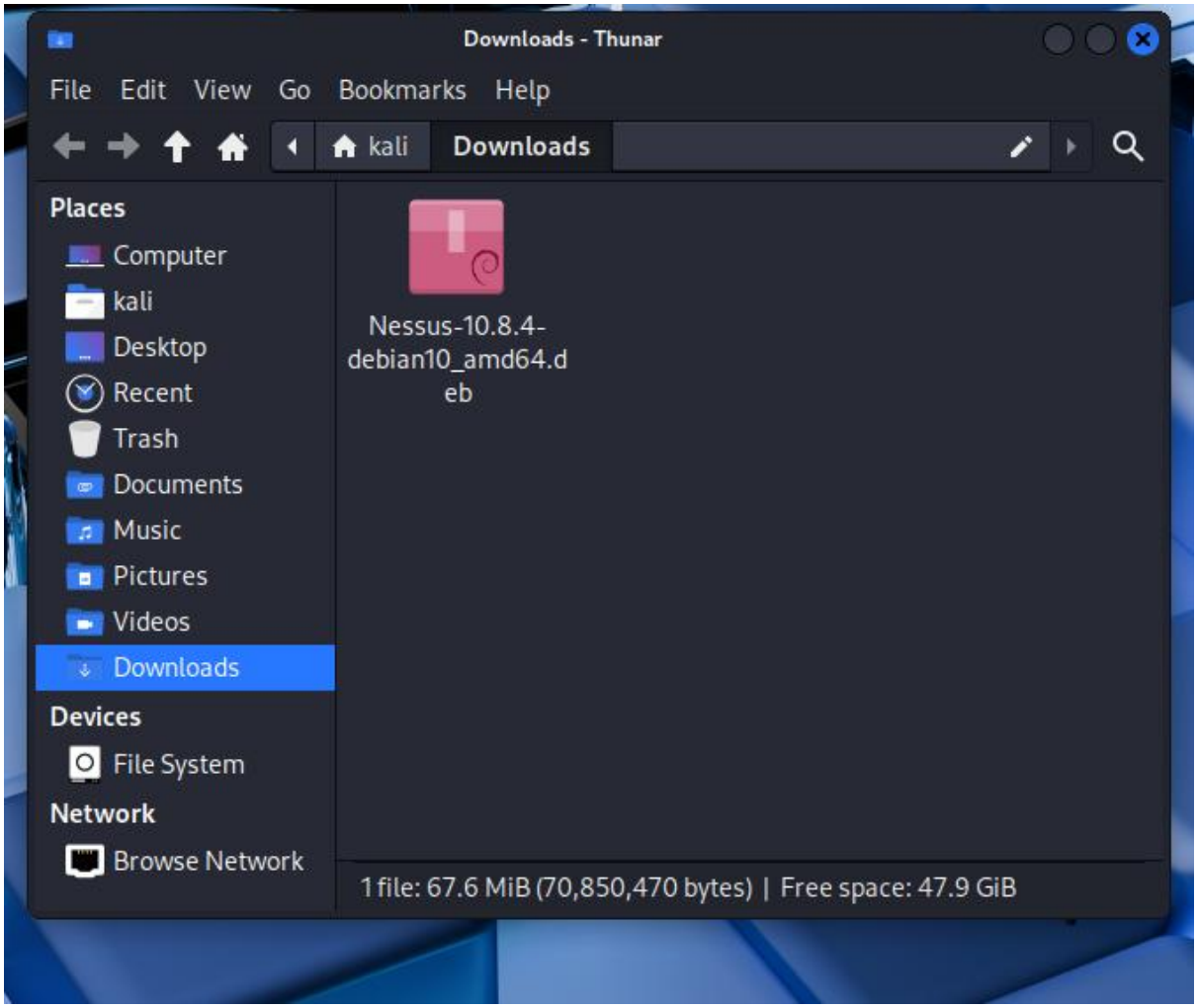


Figure: Nessus — installed via Firefox and terminal

6. Implementation Methodology

The project followed a structured five-phase approach to ensure orderly deployment and measurable progress.

Phase 1 — Planning & Design

- Documented hardware requirements and validated host capacity for target VM count.
- Designed network topology and IP addressing scheme (Appendix A).
- Produced dependency map and implementation schedule to sequence deployments correctly.

Phase 2 — Core Infrastructure

- Configured VMware Workstation Pro with four isolated virtual networks.
- Deployed and hardened pfSense; validated inter-zone routing and NAT.
- Installed Windows Server 2019; promoted to domain controller; configured AD DS and DNS.

Phase 3 — Security Infrastructure

- Deployed Security Onion with SPAN-mirrored traffic from pfSense.
- Installed Splunk Enterprise; configured Universal Forwarders on all hosts.
- Deployed Wazuh manager and agents; tuned detection rules for lab baselines.
- Implemented Suricata IPS rules and validated blocking behaviour.

Phase 4 — Endpoint & Server Deployment

- Built Windows 10 Enterprise and Ubuntu Desktop workstation templates; domain-joined Windows clients.
- Deployed Ubuntu Server for web (Apache) and mail (Postfix) services in the DMZ.
- Established internal Certificate Authority; issued SSL/TLS certificates to all services.

Phase 5 — Testing & Validation

- Conducted Nessus credentialed scans; remediated critical and high findings.
- Executed controlled penetration tests from Kali; verified IDS/SIEM detection.
- Tested all firewall rules; confirmed zero-trust policy blocked unauthorised flows.
- Validated log collection pipeline end-to-end; confirmed alert generation and notification.

7. Challenges & Resolutions

Challenge	Impact	Resolution
Hardware resource contention when running 8+ VMs concurrently	VM slowdowns; snapshot operations delayed	Staggered VM start-up order; memory balloon tuning; only run required VMs simultaneously
Complex inter-zone routing with security enforcement	Initial misrouting caused false IDS alerts	Rebuilt routing tables methodically; used pfSense packet capture to trace each flow
Linux clients failing AD authentication	Prevented domain-joined Linux workstations	Deployed realmd + SSSD; corrected DNS and Kerberos configuration
SIEM storage exhausted by log volume	Splunk indexer paused; alert gap	Implemented log rotation; source-based filtering; added 500 GB index volume
SSL certificate sprawl across multiple services	Expired certs caused service outages	Deployed internal CA; automated renewal scripts; implemented certificate inventory

8. Lessons Learned

Technical

- Defence in Depth: No single control is sufficient. Layering firewall, IDS, HIDS, and SIEM dramatically increases detection coverage.
- Logging Strategy: Centralised logging is only valuable when paired with tuned correlation rules. Raw log volume without structure creates noise, not insight.
- Identity as a Security Perimeter: Properly structured AD with Group Policy is one of the most impactful security controls in a Windows environment.
- Automation Reduces Risk: Scripted certificate renewal and VM provisioning eliminated entire categories of human-error incidents.

Professional

- Documentation First: Maintaining an up-to-date network diagram and runbook halved troubleshooting time across all phases.
- Incremental Delivery: Building and validating one layer before adding the next prevented compounding failures.
- Realistic Scoping: Complex security configurations consistently exceeded initial time estimates; buffer time is essential.

9. Technology Stack

Technology	Version	Role
VMware Workstation Pro	17.0.2	Hypervisor / virtualisation platform
pfSense	2.7.0	Firewall, NAT, VPN, DHCP
Windows Server 2019	1809	Domain controller, DNS, file services
Windows 10 Enterprise	21H2	Corporate workstations
Ubuntu Server	22.04 LTS	Web server (Apache), mail server (Postfix)
Ubuntu Desktop	22.04 LTS	Linux workstations
Security Onion	2.4	IDS/IPS, NSM, full-packet capture
Splunk Enterprise	9.0.4	SIEM — log aggregation, alerting
Wazuh	Latest	HIDS, FIM, compliance monitoring
Kali Linux	2023.1	Penetration testing & adversary simulation
Nessus Essentials	Latest	Vulnerability scanning
Fedora	18.0.2	Email server

10. Recommendations & Future Roadmap

Immediate Improvements

- Hardware Upgrade: Additional 32 GB RAM and a dedicated NVMe drive for security VMs would eliminate resource contention.
- High Availability: Implement pfSense HA with CARP for firewall redundancy.
- Automated Provisioning: Adopt Ansible playbooks for repeatable, version-controlled VM configuration.

Medium-Term Expansion

- Cloud Hybrid Integration: Extend the lab into AWS or Azure to simulate hybrid-cloud attack paths.
- ML-Based Anomaly Detection: Integrate an UEBA tool (e.g., Elastic ML) to detect behavioural anomalies.
- Compliance Benchmarking: Map controls to NIST CSF and CIS Controls; produce gap analysis reports.

Educational Goals

- Develop structured red/blue team exercise scenarios with defined success criteria.
- Create incident response playbooks for common attack patterns (ransomware, credential theft, C2).

- Publish lab configurations as Infrastructure-as-Code (Terraform + Ansible) for community sharing.

11. Conclusion

This cybersecurity homelab represents a significant practical achievement: the design and delivery of a fully functional, multi-segment enterprise security environment. From architecture planning through to adversary simulation, every phase was guided by industry best practices and real-world operational considerations.

The skills demonstrated — network design, firewall policy, identity management, SIEM engineering, intrusion detection, and vulnerability management — map directly onto the competencies demanded by cybersecurity employers. More importantly, the lab is a living platform: it will continue to evolve with new attack scenarios, tools, and compliance benchmarks.

This project is not just an academic exercise. It is evidence of the ability to plan, build, secure, and operate a production-grade network environment independently.

Appendix A — IP Addressing Scheme

Zone	CIDR	Gateway	Hosted Services
Management	10.10.10.0/24	10.10.10.1	Admin consoles, OOB access
Corporate LAN	192.168.1.0/24	192.168.1.1	AD, DNS, workstations, file server
DMZ	172.16.1.0/24	172.16.1.1	Apache web, Postfix mail, Nginx proxy
Security VLAN	10.0.0.0/24	10.0.0.1	Security Onion, Splunk, Wazuh, Kali

Appendix B — Learning Resources

- Cybercademy — Cybersecurity Homelab Project Guide
- Chris Long — Detection Lab documentation
- Lawrence Systems — pfSense configuration video series
- Microsoft Learn — AD DS and Windows Server documentation
- Security Onion — Official documentation and community forums
- Splunk — Official documentation and Splunk University free courses

- Reddit r/homelab and r/netsec communities
- NIST SP 800-53 — Security and Privacy Controls for Information Systems